Turbidity Bridge Comparison for PA

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Methods

Download Turbidity Data from CDEC (15-minute data)

Clean turbidity data

Remove dates with less than 12 values in a day

```
turb_df <- readRDS("data_raw/turbidity_cdec_OBI_HOL_OH4_OSJ_2010-2023.rds")

turb_qc <- turb_df %>%
  mutate(hour = hour(datetime)) %>%
  group_by(station, wy, month, date, hour) %>%
  slice(1) %>%
  ungroup() %>%
  select(-datetime) %>%
  group_by(station, wy, month, date) %>%
  summarise(total = n()) %>%
  mutate(Flag_QC = ifelse(total < 13, "Y", "N"))  %>%
  select(-total)

turb_q1 <- turb_df %>% left_join(turb_qc) %>%
  filter(Flag_QC == "N") %>%
  select(datetime,date,wy, month, station , turbidity)
```

Remove periods where values are repetitive (likely sensor issue)

```
repeating_vals = function(df, x){
  df$same = ifelse(df$turbidity == lag(df$turbidity, 1, default = 0), 1L, 0L)
  df = df %>%
    mutate(issame = cumsum(df$same == 0L)) %>%
    group_by(station, issame) %>%
    mutate(flag = sum(same)+1 ) %>%
    ungroup() %>%
    mutate(Flag_repeats = ifelse(flag > x, "Y", "N"))
  return(df)
}
```

threshold = 6 hours in a row (24 points)

```
# Run function repeating values and get rid of the columns we don't need
turb_q2 <- repeating_vals(df = turb_q1, x = 24) %>%
  select(-flag, -issame, -same) %>%
  rename(Flag_QC2 = Flag_repeats)

turb_q2_clean <- turb_q2 %>%
  filter(Flag_QC2 == "N")

# Flagged values
turb_flagged <- turb_q2_clean %>%
  filter(Flag_QC2 == "Y")
```

Remove high values (>=350 FNU)

```
turb_clean <- turb_q2_clean %>%
  filter(turbidity<350)

# saveRDS(turb_clean, "data_clean/clean_turbidity_OBI_HOL_OH4_OSJ_event.rds")
```

Look at data

```
turb_clean <- readRDS(here("data_clean/clean_turbidity_OBI_HOL_OH4_OSJ_event.rds"))

# png("figures/HOL_OBI_OH4_OSJ_lightlycleaneddata.png", res = 300, units = "in", width = 6, height = 4)
#
# ggplot(turb_clean) + geom_point(aes(date, turbidity)) + facet_wrap(~station)
#
# dev.off()

knitr::include_graphics("figures/HOL_OBI_OH4_OSJ_lightlycleaneddata.png")
```

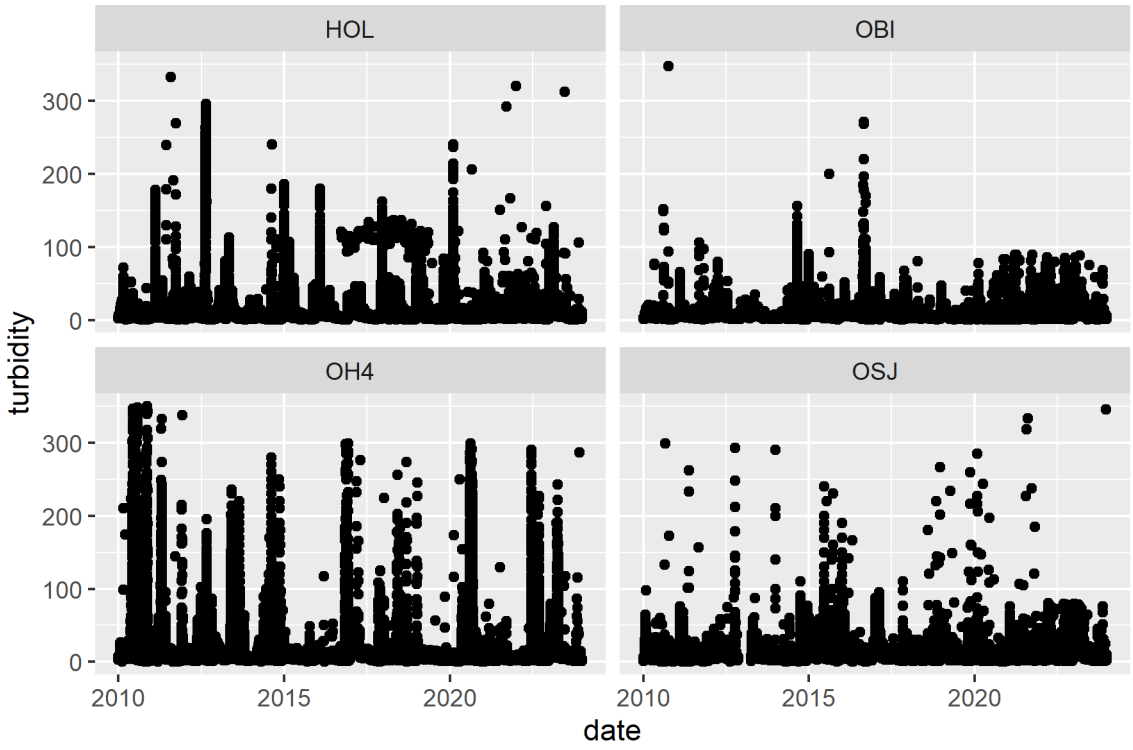


Figure 1: Event (15-minute) data from CDEC for HOL, OBI, OH4, OSJ after cleaning (see methods for steps).

Data are still messy with light cleaning.

Convert to daily data

- Convert to daily
- Filter to December-March

```
turb_daily <- turb_clean %>%
  group_by(date, month, wy, station) %>%
  summarize(turbidity_daily = mean(turbidity)) %>%
  ungroup() %>%
  mutate(date2 = paste0(1980, "-", month, "-", day(date)),
         date2 = lubridate::ymd(date2),
         year = year(date),
         fWY = factor(wy)) %>%
  filter(month %in% c(12, 1, 2, 3))
```

Calculate number of days under each option where Turbidity Bridge would be triggered

```
turb_wide <- turb_daily %>%
  pivot_wider(names_from = "station",
```

```
              values_from = "turbidity_daily") %>%
  mutate(Opt1 = if_else(OBI >= 12, 1L, 0L),
         Opt2 = if_else(OBI >=12 & HOL >=12 & OH4 >=12, 1L, 0L),
         Opt3 = if_else(OBI >=12 & HOL >=12 & OSJ >=12, 1L, 0L))

turb_days <- turb_wide %>%
  select(wy, fWY, Opt1, Opt2, Opt3) %>%
  pivot_longer(cols =c(Opt1, Opt2, Opt3), names_to = "Option", values_to = "exceeding_threshold") %>%
  group_by(wy, fWY, Option) %>%
  summarize(days_exceeding = sum(exceeding_threshold,na.rm = TRUE)) %>%
  ungroup() %>%
  mutate(Option = case_when(Option == "Opt1" ~ "OBI only" ,
                    Option == "Opt2" ~ "OBI, HOL, OH4",
                    Option == "Opt3" ~ "OBI, HOL, OSJ"))
```

Results

Plots and Tables

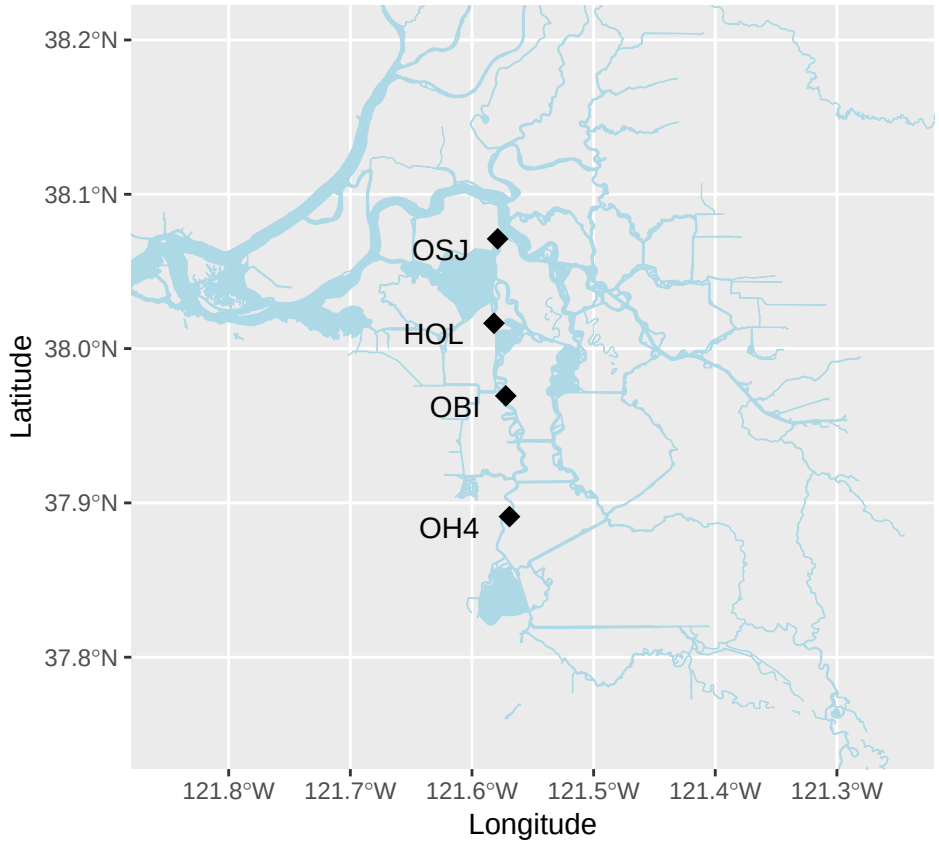


Figure 2: Map of Turbidity Stations

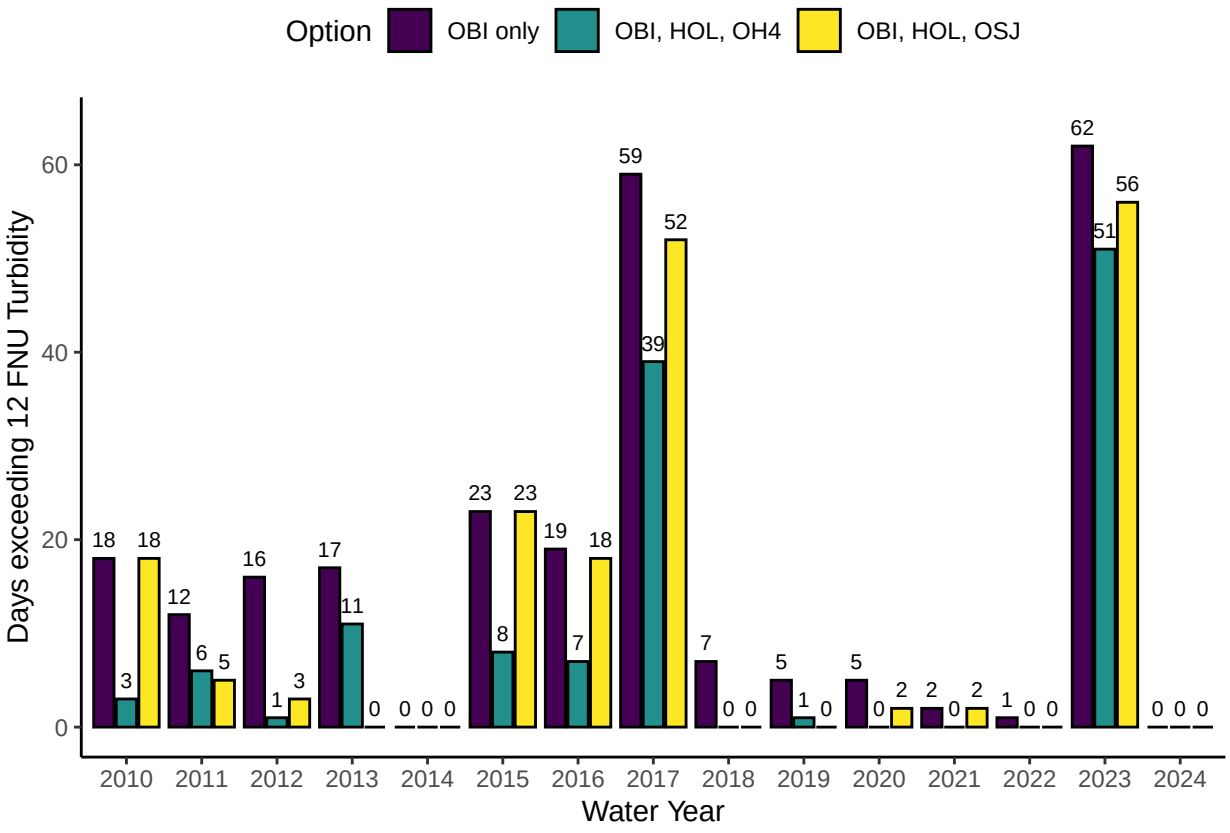


Figure 3: Days Triggering Turbidity Bridge Action for Different Turbidity Station Combinations.

Table 1: Adult Delta Smelt Turbidity Bridge Action: Days meeting trigger under various combinations of stations.

wy	OBI only	OBI, HOL, OH4	OBI, HOL, OSJ
2010	18	3	18
2011	12	6	5
2012	16	1	3
2013	17	11	0
2014	0	0	0
2015	23	8	23
2016	19	7	18
2017	59	39	52
2018	7	0	0
2019	5	1	0
2020	5	0	2
2021	2	0	2
2022	1	0	0
2023	62	51	56
2024	0	0	0

Notes

- High flow offramp specifications are not considered in this analysis.
- Dates are filtered to Dec 1 - March 31 of each year to represent the season for Adult Delta Smelt. RVB and SJJ have reached 3-day average water temperatures of 12°C/53.6°F (representing a shift to larval Delta Smelt lifestage) prior to April 1 since 2009 (Figure 4). First detection of Delta Smelt larvae also generally occurs prior to April 1.

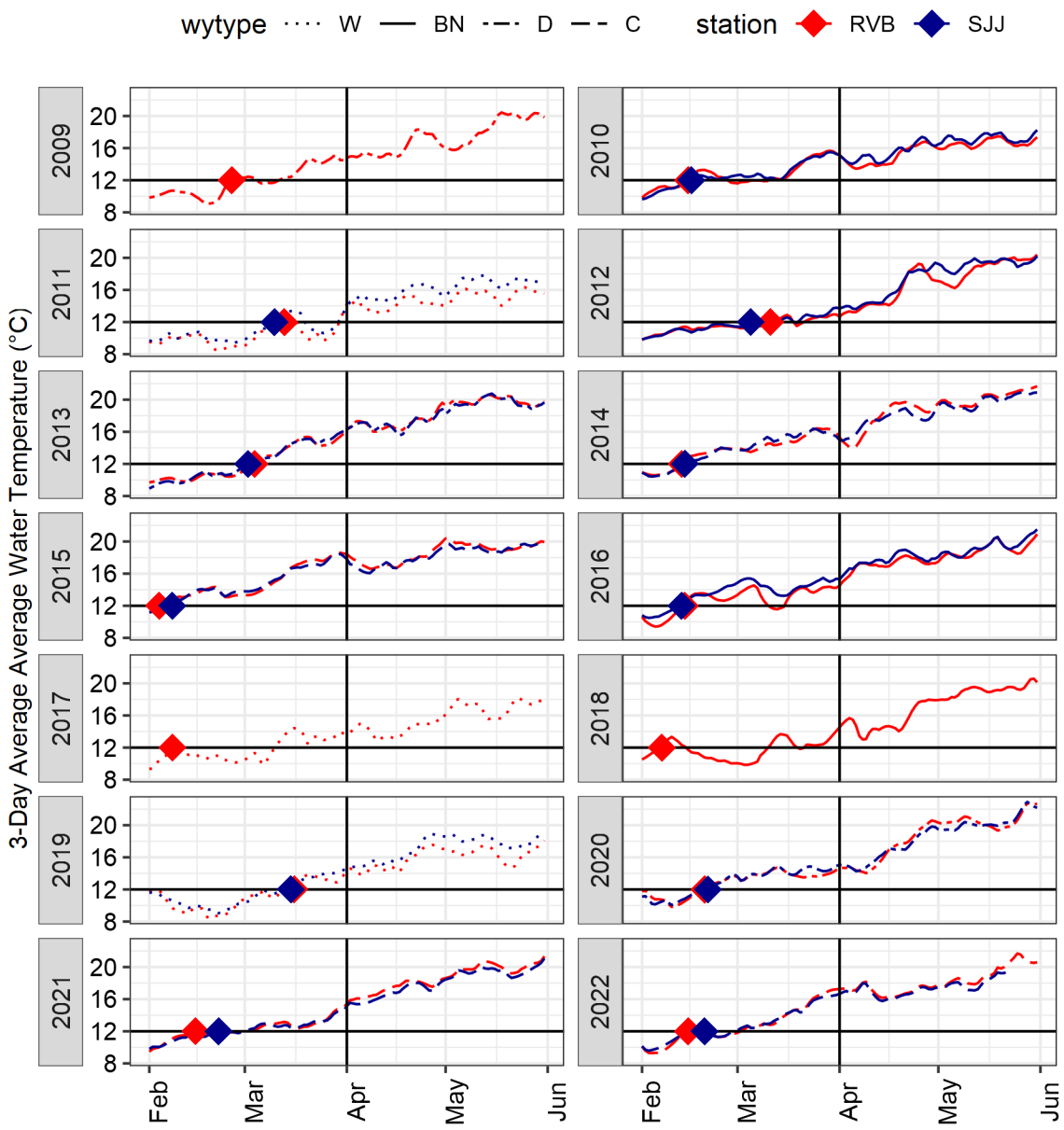


Figure 4: 3-day Average Water Temperatures at Rio Vista Bridge (RVB) and San Joaquin River at Jersey Point (SJJ), WY 2009-2021. Certain years are missing SJJ data (2009, 2017, 2018). Line pattern indicates water year type.